GIRIK SHAILENDRA TRIPATHI

San Francisco, CA (Open to Relocate) | Email | (240) 462-1347 | LinkedIn | GitHub | Portfolio

SUMMARY

AI Software Engineer with 4+ years of experience designing scalable backend architectures, high-throughput data pipelines, and production-grade AI/LLM systems on AWS. Proven track record of architecting event-driven microservices, RAG pipelines, and vector search infrastructure that process terabyte-scale data with high availability and sub-100ms P99 latencies. Expertise in Python, Go, PyTorch, LangChain, Vector Databases (pgvector, Pinecone, Qdrant), REST/gRPC APIs, and distributed cloud computing. M.S. in Data Science.

SKILLS

AI & GenAI Engineering: LangChain, LlamaIndex, PyTorch, Retrieval-Augmented Generation, Agentic Workflows, Vector Search, pgvector, Qdrant, vLLM, FastAPI, LangSmith, Ragas
Backend Architecture & Systems: Python, Java, Go, SQL, PostgreSQL, MySQL, REST APIs, gRPC, Microservices, Event-Driven Systems, Distributed Systems, Asynchronous Queues, API Gateway
Cloud & Infrastructure: AWS Lambda, AWS ECS, DynamoDB, AWS S3, AWS SQS, Azure OpenAI, Azure Functions, AKS, Docker, Kubernetes, Terraform
Data Engineering: Apache Spark, PySpark, Apache Kafka, Airflow, dbt, ETL Data Pipelines, Schema Design, Data Modeling
Testing & Quality Assurance: pytest, unittest, Postman API Testing, Integration Testing, End-to-End Testing, Mocking, Locust Load Testing, LLM Evaluation
DevOps & Observability: CI/CD, AWS CodePipeline, Azure DevOps, Jenkins, CloudWatch, Prometheus, Grafana, IAM, Linux Shell Scripting, Git, JIRA, Agile

EXPERIENCE

DoorDash, United States | *AI Software Engineer* **January 2026 – Present**

- Architected an asynchronous agentic routing service using Python, LangChain, and AWS ECS, automating merchant catalog classification and reducing menu ingestion delays by 38% across 50,000+ active partner profiles.
- Pioneered a hybrid vector search pipeline leveraging pgvector, Qdrant, and Redis caching to power natural-language inventory search, achieving sub-65ms P99 latency during peak load periods exceeding 12,000 requests/sec.
- Established an automated LLM evaluation framework using pytest, LangSmith, and Ragas, setting regression testing guardrails across fine-tuned inference pipelines and cutting support hallucination rates by 24%.
- Optimized real-time dispatch telemetry microservices running on AWS Lambda, SQS, and DynamoDB, integrating low-latency AI inference endpoints to predict order preparation times and improving delivery ETA accuracy by 15%.
- Streamlined automated CI/CD and observability pipelines via AWS CodePipeline, Docker, and CloudWatch, enabling continuous model evaluation deployments with zero-downtime rollbacks and reducing incident detection times by 45%.

Accenture, India | *Software Development Engineer* **October 2022 – August 2024**

- Engineered high-throughput event-driven microservices using Java, Go, gRPC, and AWS Lambda, replacing legacy monolith endpoints for a Fortune 500 client and boosting API throughput by 2.5x under peak concurrency.
- Constructed asynchronous data streaming pipelines using Apache Kafka and PySpark on AWS EMR, processing 8 TB+/month of transaction records with automated schema validation and cutting nightly ETL processing windows by 40%.
- Spearheaded automated document processing and NLP extraction workflows using PyTorch, Hugging Face Transformers, and LlamaIndex to parse unstructured vendor invoices, reducing manual data-reconciliation effort by 60%.
- Executed robust unit and integration testing suites using pytest, Postman API testing, and Locust, achieving 88% code coverage and identifying capacity bottlenecks prior to production releases.
- Automated enterprise CI/CD and infrastructure-as-code pipelines using Jenkins, Terraform, and AWS CloudWatch, accelerating multi-environment deployments and reducing infrastructure setup time from 3 days to under 45 minutes.

Aspire Technolab, India | *Software Engineer* **January 2021 – September 2022**

- Developed RESTful microservices and backend API modules using Python, PostgreSQL, and MySQL, hosted on Azure App Services and improving API response times by 32% through query optimization.
- Structured normalized relational database schemas governing 100,000+ transactional records, establishing foreign key constraints and automated data validation scripts that reduced data consistency errors by 45%.
- Created automated data ingestion scripts using Python, SQL, and Linux shell commands to parse, clean, and map multi-source CSV and JSON payloads into centralized database tables, cutting manual processing time by 50%.
- Implemented automated unit and integration tests using unittest and Postman, expanding test suite coverage to 82% and reducing post-release defect reports across production client builds.
- Configured containerized build and deployment pipelines using Docker, Git, and Azure DevOps, streamlining developer onboarding and standardizing CI/CD release cycles.

EDUCATION

Master of Science in Data Science, *University of Maryland, College Park, United States* **May 2026**
BE in Electronics & Telecommunication, *University of Mumbai, India* **June 2022**

PROJECTS

Crisis Pulse – Real-Time AI Threat & Event Intelligence Platform **December 2025 – February 2026**

- Architected an event-streaming AI backend using AWS Kinesis, Lambda, DynamoDB, and Qdrant to ingest 10,000+ events/sec, using a RAG synthesis pipeline including FastAPI, LangChain, pgvector with hybrid search (BM25 + dense vectors) to generate sub-100ms geo-tagged threat summaries.
- Engineered agentic alerting and evaluation guardrails using EventBridge, SNS, and structured output parsing (Pydantic), integrating pytest and Ragas to eliminate hallucinations and ensure fault-tolerant, zero-message-loss incident delivery.